Table of Contents

FOUNDATIONS

Chapter 1. The Breadth and Depth of DSP 1

The Roots of DSP 1
Telecommunications 4
Audio Processing 5
Echo Location 7
Imaging Processing 9

Chapter 2. Statistics, Probability and Noise 11

Signal and Graph Terminology 11
Mean and Standard Deviation 13
Signal vs. Underlying Process 17
The Histogram, Pmf and Pdf 19
The Normal Distribution 26
Digital Noise Generation 29
Precision and Accuracy 32

Chapter 3. ADC and DAC 35

Quantization 35
The Sampling Theorem 39
Digital-to-Analog Conversion 44
Analog Filters for Data Conversion 48
Selecting the Antialias Filter 55
Multirate Data Conversion 58
Single Bit Data Conversion 60

Chapter 4. DSP Software 67

Computer Numbers 67
Fixed Point (Integers) 68
Floating Point (Real Numbers) 70
Number Precision 72
Execution Speed: Program Language 76
Execution Speed: Hardware 80
Execution Speed: Programming Tips 84